



Acme Gourmet Meals Takeoff

DATASCI 205
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Introduction

1. AGM's Future Vision

- more pickup locations
- using BART system
- alternate methods of transportation

2. NoSQL Graph Databases – Business Cases

- neo4j
- mongoDB
- redis





**Louvain Modularity
+
Geodesic Fencing**



**Harmonic
Centrality**

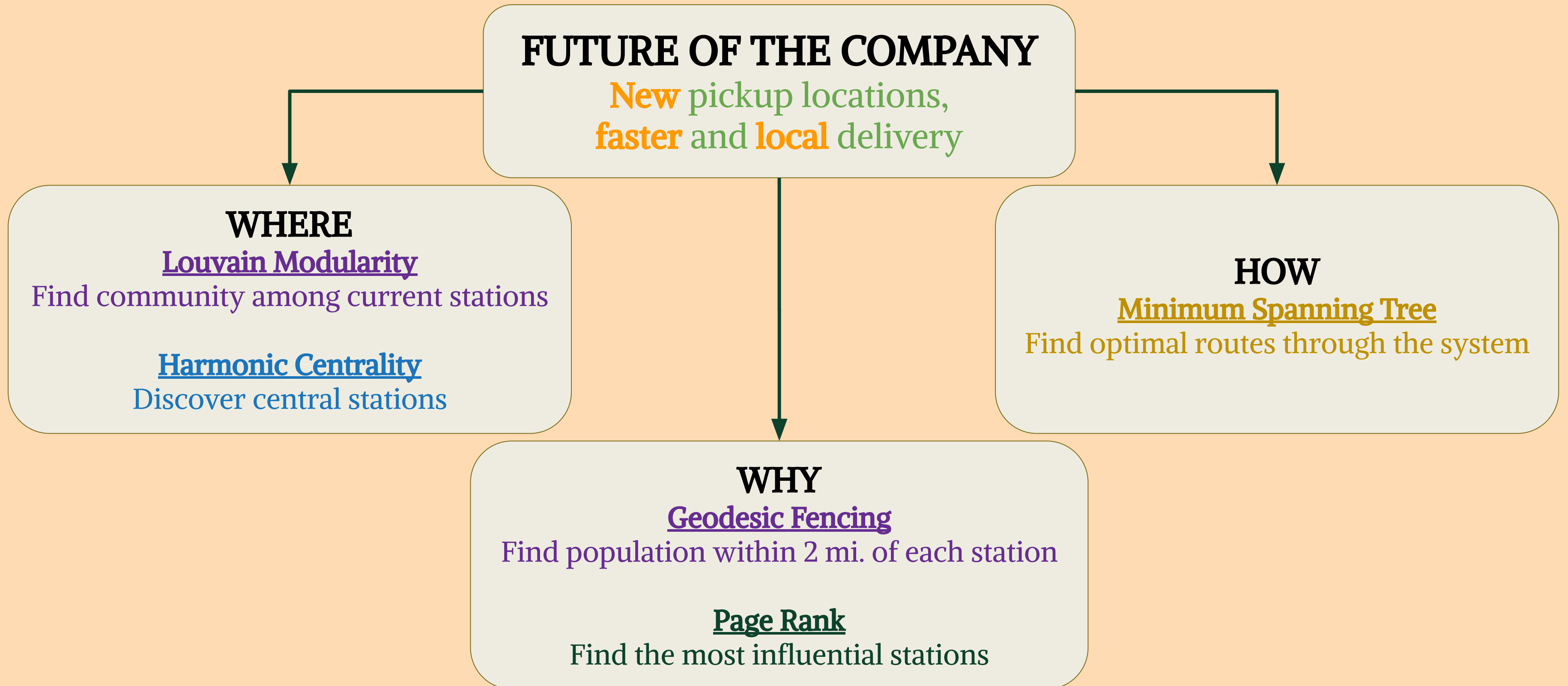


PageRank



**Minimum
Spanning Tree**

AGM Business Cases – Neo4j





Louvain Modularity + Geodesic Fencing

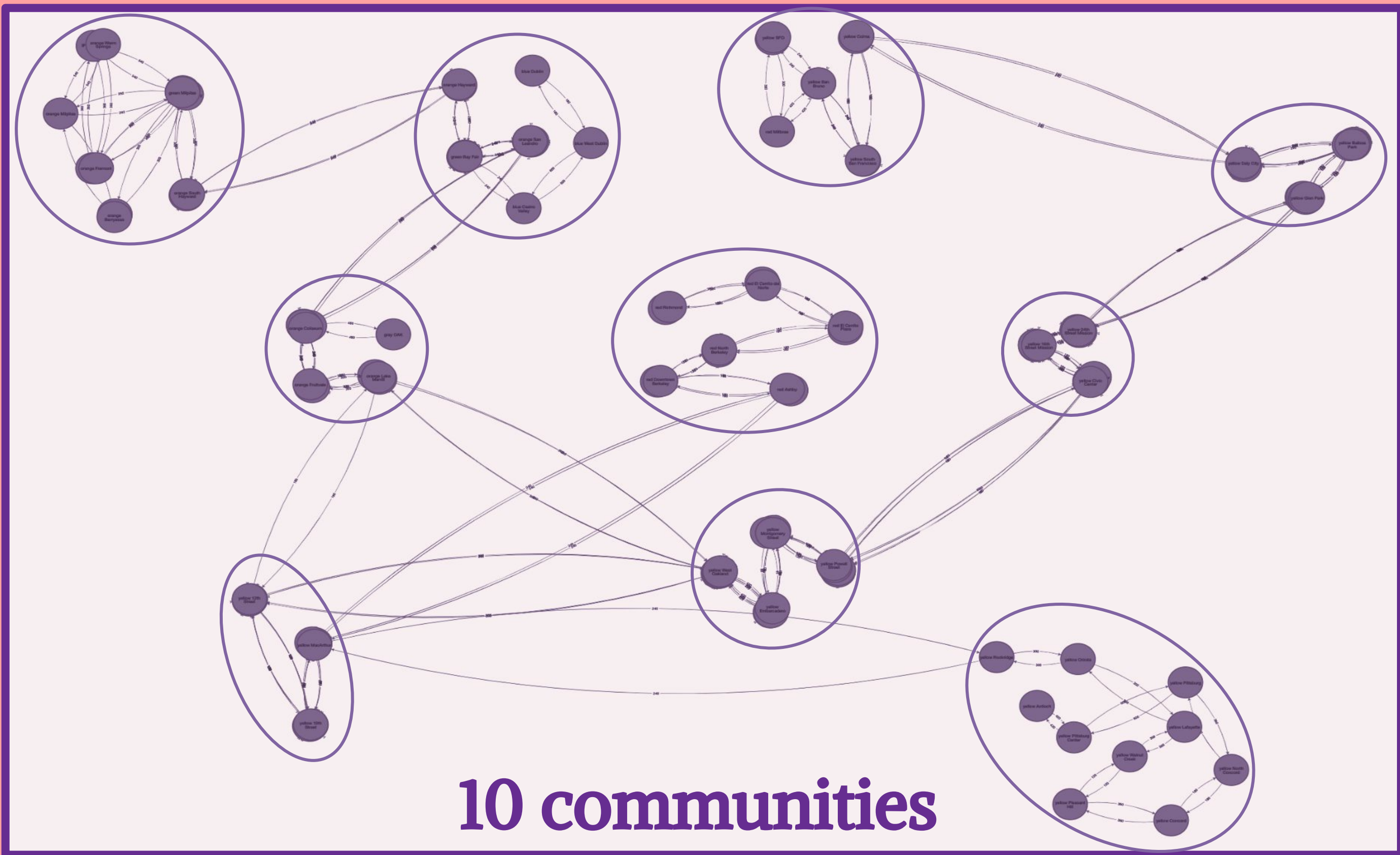
Louvain Modularity – Result Characteristics

	name_of_code	community	population
0	Antioch	52	66933
1	Concord	52	87117
2	Lafayette	52	29639
3	North Concord	52	58689
4	Orinda	52	19341
5	Pittsburg	52	0
6	Pittsburg Center	52	96081
7	Pleasant Hill	52	86258
8	Rockridge	52	151725
9	Walnut Creek	52	43556
10	Balboa Park	54	253123
11	Daly City	54	231056
12	Glen Park	54	287877
22	Bay Fair	67	93041
23	Castro Valley	67	110328
24	Dublin	67	54420

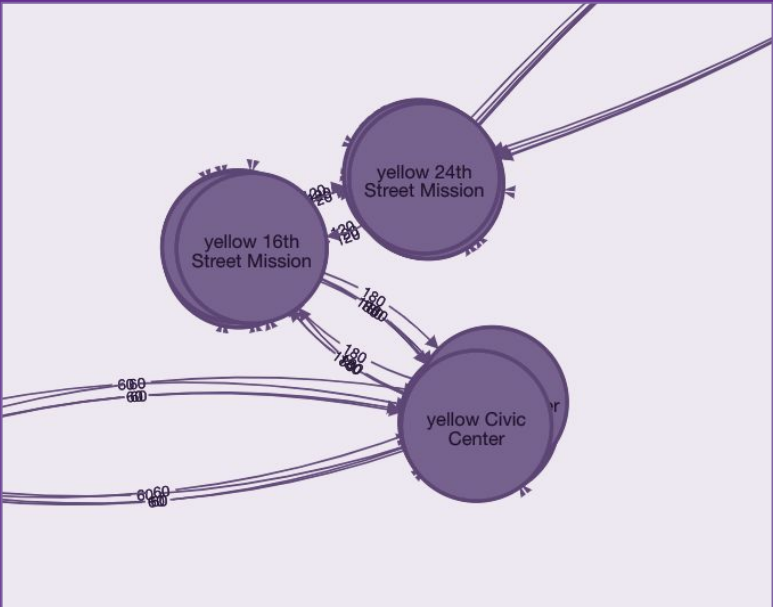
25	San Leandro	67	123740
26	West Dublin	67	54420
28	Hayward	67	66056
33	Ashby	74	173897
34	Downtown Berkeley	74	153753
35	El Cerrito Plaza	74	70450
36	El Cerrito del Norte	74	81472
37	North Berkeley	74	134802
38	Richmond	74	85676
45	Colma	75	114402
46	Millbrae	75	22845
47	SFO	75	22845
48	San Bruno	75	110856
49	South San Francisco	75	186691
54	Coliseum	92	123040
55	Fruitvale	92	90602
56	Lake Merritt	92	175958

58	OAK	92	84499
65	Berryessa	93	197640
66	Fremont	93	73855
67	Milpitas	93	31170
68	South Hayward	93	93452
69	Union City	93	148456
70	Warm Springs	93	120913
77	16th Street Mission	98	339093
78	24th Street Mission	98	315201
79	Civic Center	98	347795
89	Embarcadero	106	210921
90	Montgomery Street	106	268802
91	Powell Street	106	303556
92	West Oakland	106	42316
105	12th Street	110	175958
106	19th Street	110	165215
107	MacArthur	110	143953

Louvain Modularity – Results

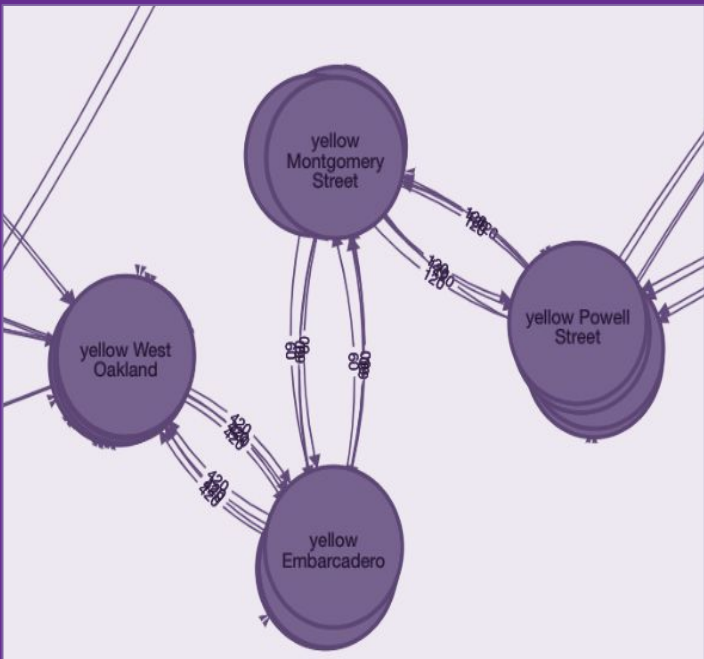


Geodesic Fencing



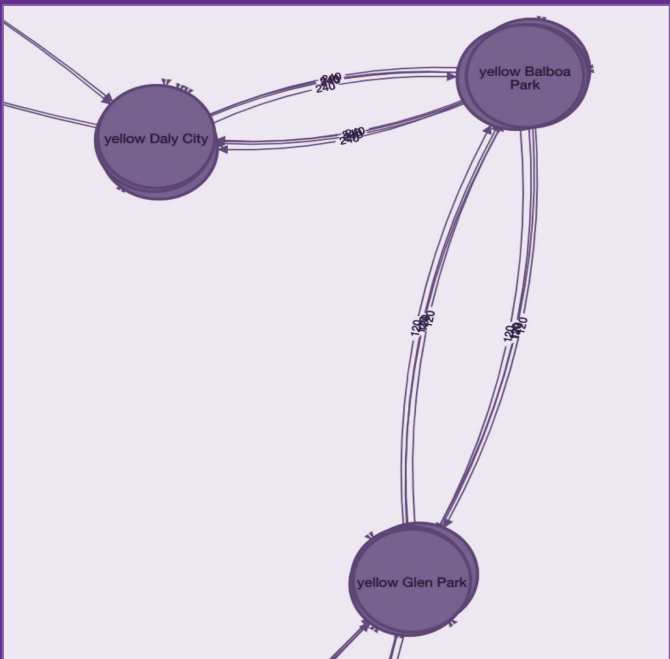
Population: 1,002,089

16th Street Mission	339,093
24th Street Mission	315,201
Civic Center	347,795



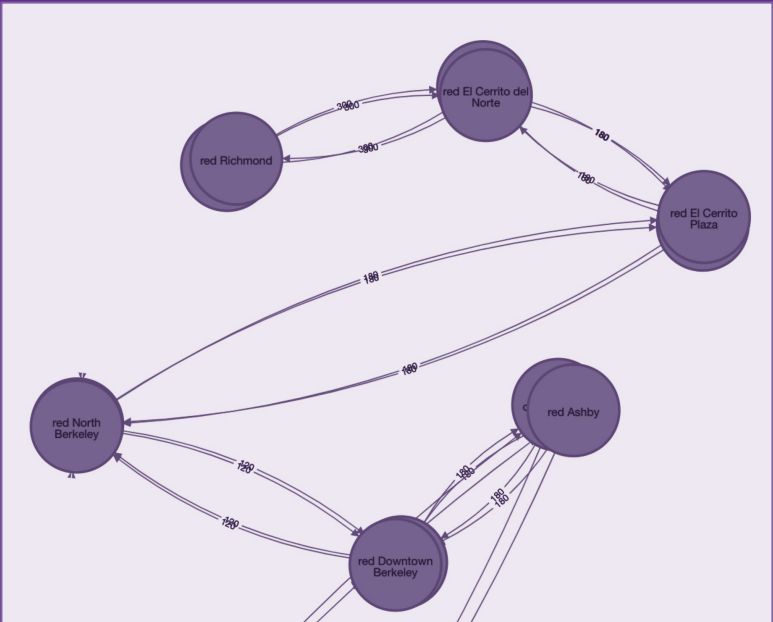
Population: 825,595

Embarcadero	210,921
Montgomery Street	268,802
Powell Street	303,556
West Oakland	42,316



Population: 772,056

Balboa Park	253,123
Daly City	231,056
Glen Park	287,877



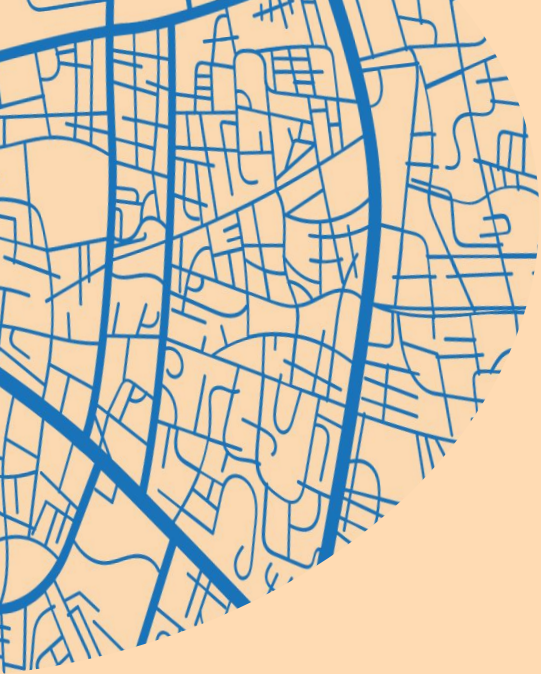
Population: 700,050

Ashby	173,897
Downtown Berkeley	153,753
El Cerrito Plaza	70,450
El Cerrito del Norte	81,472
North Berkeley	134,802
Richmond	85,676

Harmonic Centrality

- Measures shortest path between a node and all other nodes





Harmonic Centrality

“To Move or Not to Move the Current Kitchen”

Algorithm Results

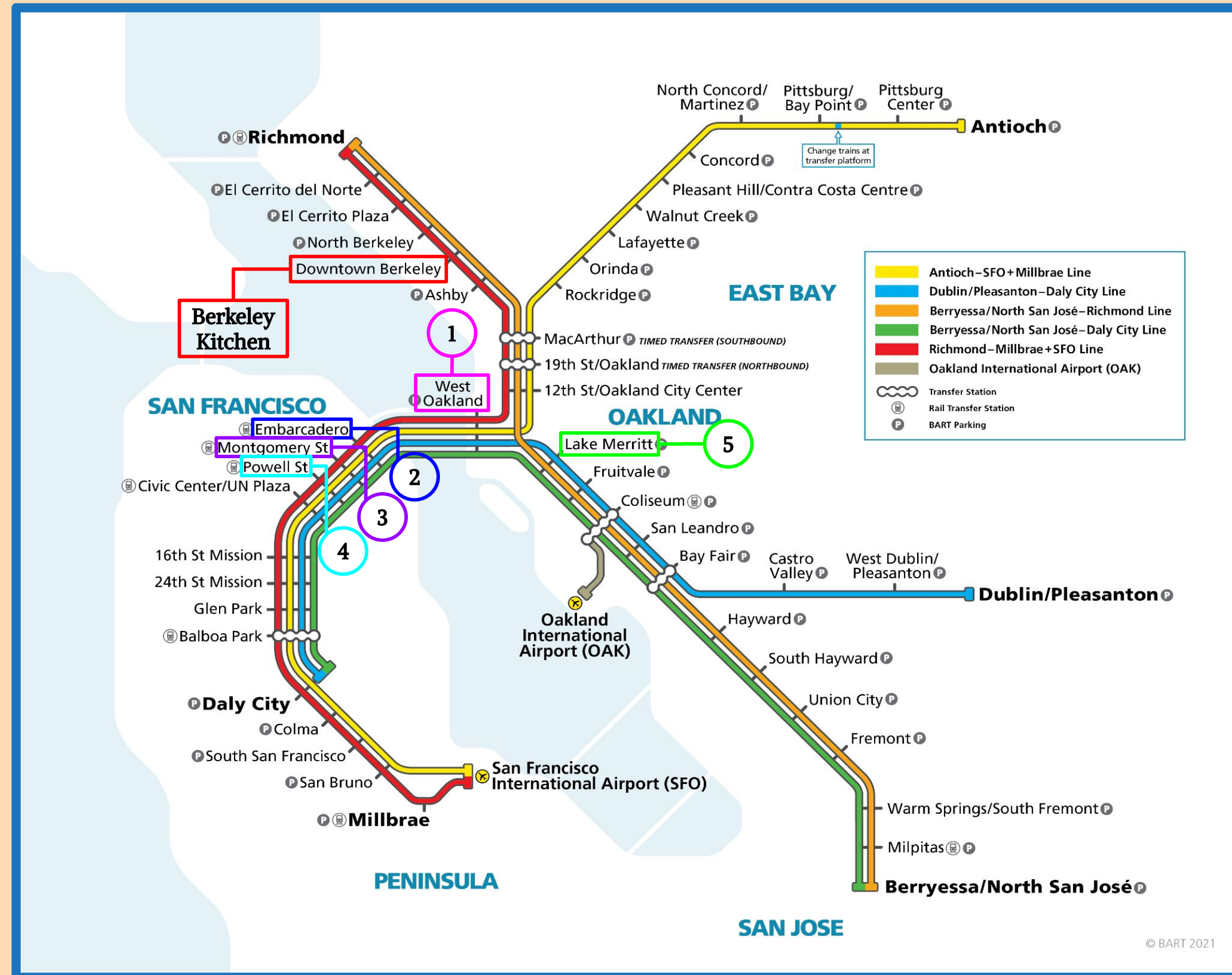
	name	closeness
0	green West Oakland	0.160198
1	yellow West Oakland	0.159922
2	blue West Oakland	0.159903
3	red West Oakland	0.159674
4	green Embarcadero	0.154619
...	...	...
209	depart Union City	0.000000
210	depart Walnut Creek	0.000000
211	depart Warm Springs	0.000000
212	depart West Dublin	0.000000
213	depart West Oakland	0.000000

214 rows × 2 columns

Consolidating Line Nodes per Station

	name	closeness
6	West Oakland	0.131206
21	Embarcadero	0.126800
28	Montgomery Street	0.123487
26	Powell Street	0.120650
45	Lake Merritt	0.120233

Harmonic Centrality – BART System





PageRank

- Most influential stations

PageRank

- 4 stations on X-shaped BART graph
 - Pre-PageRank: distribution point in the center
 - Post-PageRank: distribution points on influential stations

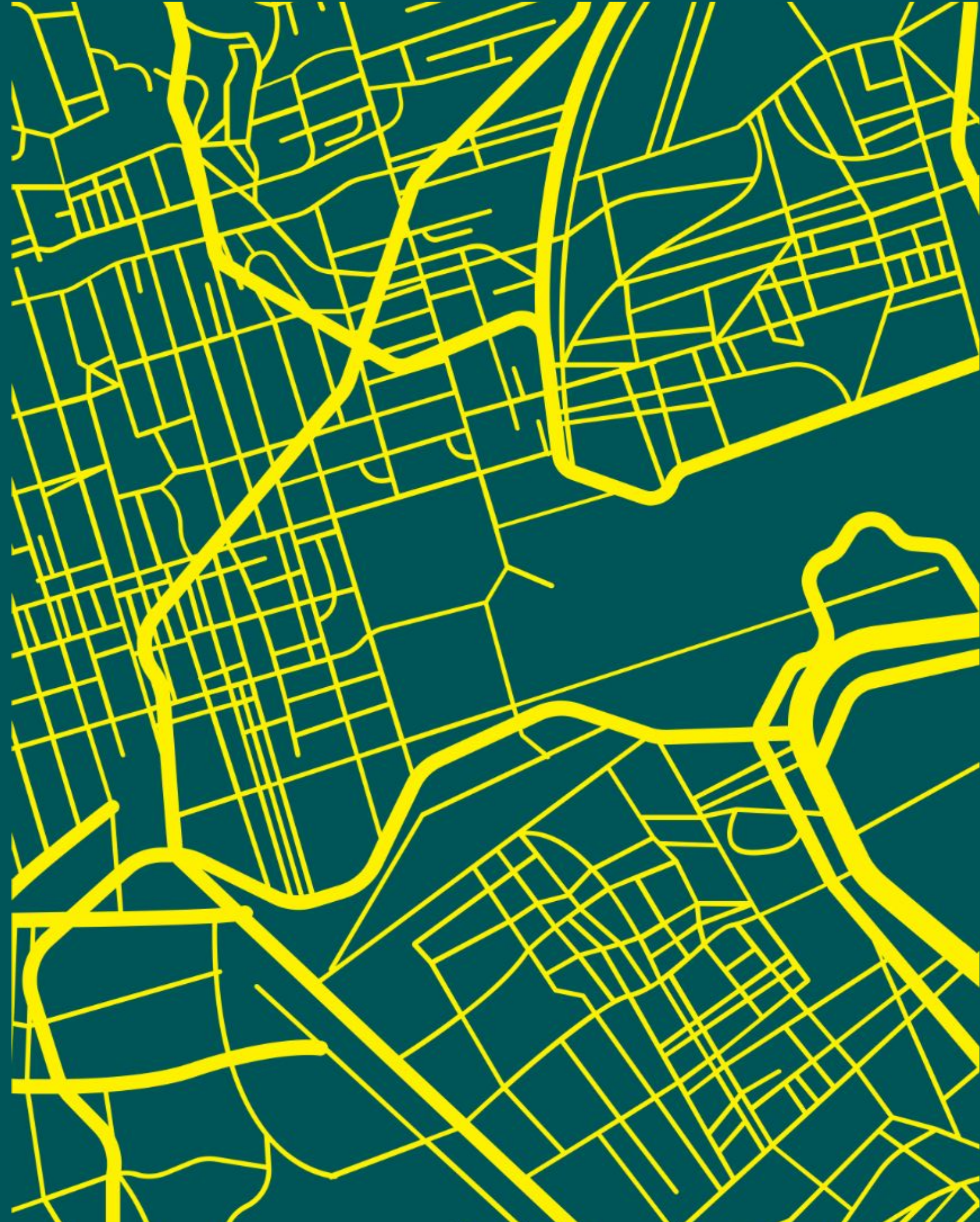


	name	page_rank	line	sequence
0	Coliseum	0.690175	blue	6
1	MacArthur	0.675697	yellow	11
2	Daly City	0.674432	red	19
3	Pittsburg	0.645705	yellow	3

PageRank – BART System



Minimum Spanning Tree (MST)



Minimum Spanning Tree – Uses

- **BART**

- **Drones** – pick up at station and deliver locally
- **Robots** – pick up at station and deliver locally

- **Not BART**

- **Drones** – pick up at additional center and deliver locally
- **Robots** – pick up at additional center and deliver locally

Minimum Spanning Tree

1. Zip codes within 2 mi. of Downtown Berkeley
2. Nodes and edges of graph
3. Distance between nodes
 - a. edge weight = distance



Minimum Spanning Tree – Nodes, Edges, Results

Nodes :		
	node_name	labels
0	arrive 94618	[Station]
1	arrive 94702	[Station]
2	arrive 94703	[Station]
3	arrive 94704	[Station]
4	arrive 94705	[Station]
5	arrive 94706	[Station]
6	arrive 94707	[Station]
7	arrive 94709	[Station]
8	arrive 94710	[Station]
9	arrive 94720	[Station]
10	arrive Downtown Berkeley	[Station]
11	depart 94618	[Station]
12	depart 94702	[Station]
13	depart 94703	[Station]
14	depart 94704	[Station]
15	depart 94705	[Station]
16	depart 94706	[Station]
17	depart 94707	[Station]
18	depart 94709	[Station]
19	depart 94710	[Station]
20	depart 94720	[Station]
21	depart Downtown Berkeley	[Station]



Relationships:			
	node_name_1	relationship_type	node_name_2
0	arrive 94618	TRACK	depart 94618
1	arrive 94618	TRACK	depart 94618
2	arrive 94618	TRACK	depart 94702
3	arrive 94618	TRACK	depart 94703
4	arrive 94618	TRACK	depart 94704
...	...	...	...
147	depart Downtown Berkeley	TRACK	arrive 94707
148	depart Downtown Berkeley	TRACK	arrive 94709
149	depart Downtown Berkeley	TRACK	arrive 94710
150	depart Downtown Berkeley	TRACK	arrive 94720
151	depart Downtown Berkeley	TRACK	arrive Downtown Berkeley
152 rows × 5 columns			
Density: 0.7			

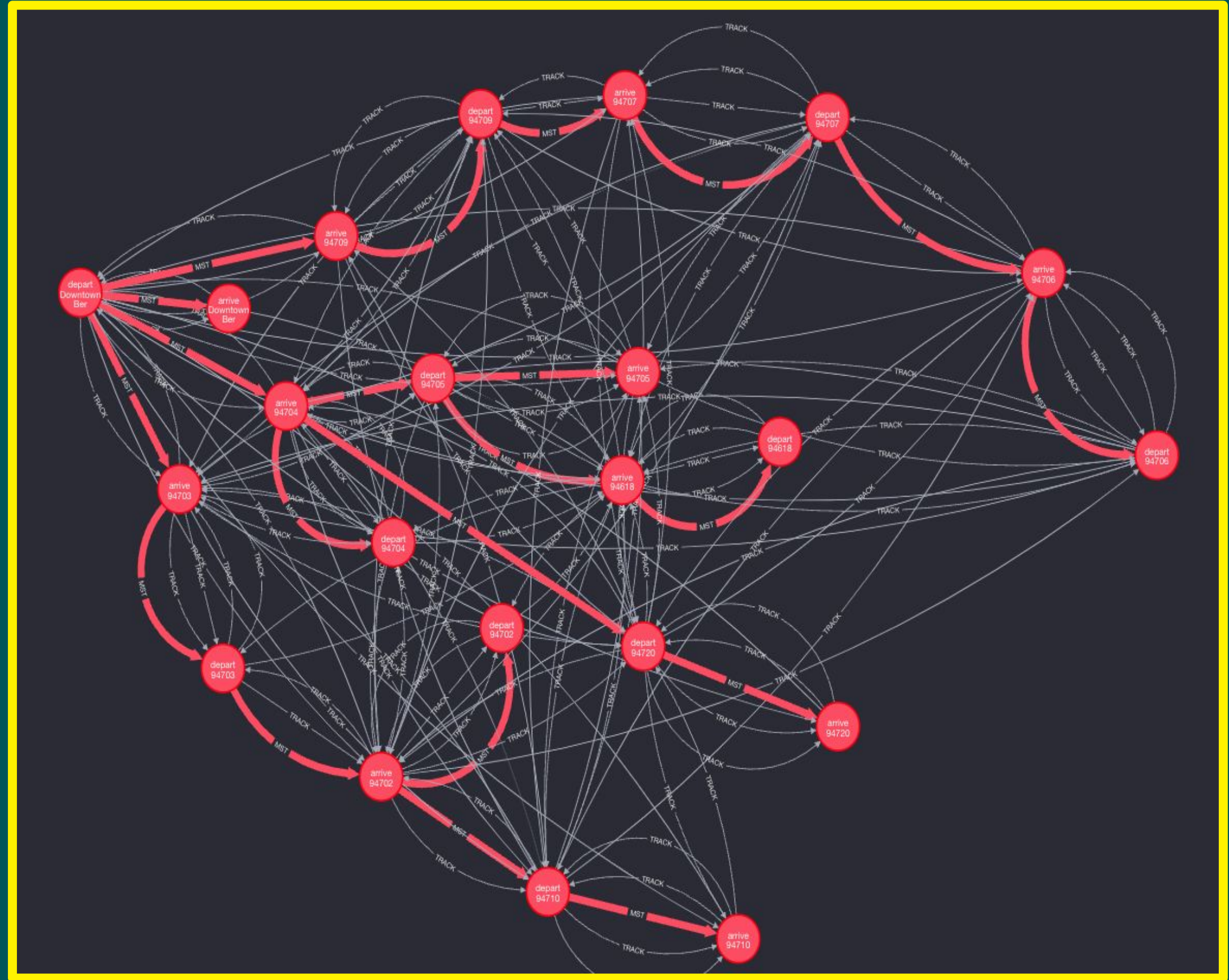


	source	destination	cost
0	depart Downtown Berkeley	arrive 94704	0.599582
1	arrive 94704	depart 94704	0.000000
2	arrive 94704	depart 94705	1.086881
3	depart 94705	arrive 94618	1.462633
4	arrive 94618	depart 94618	0.000000
5	depart 94705	arrive 94705	0.000000
6	arrive 94704	depart 94720	0.539522
7	depart 94720	arrive 94720	0.000000
8	depart Downtown Berkeley	arrive 94703	0.573894
9	arrive 94703	depart 94703	0.000000
10	depart 94703	arrive 94702	0.599537
11	arrive 94702	depart 94702	0.000000
12	arrive 94702	depart 94710	0.924355
13	depart 94710	arrive 94710	0.000000
14	depart Downtown Berkeley	arrive Downtown Berkeley	0.000000
15	depart Downtown Berkeley	arrive 94709	0.659097
16	arrive 94709	depart 94709	0.000000
17	depart 94709	arrive 94707	1.472809
18	arrive 94707	depart 94707	0.000000
19	depart 94707	arrive 94706	1.070267
20	arrive 94706	depart 94706	0.000000

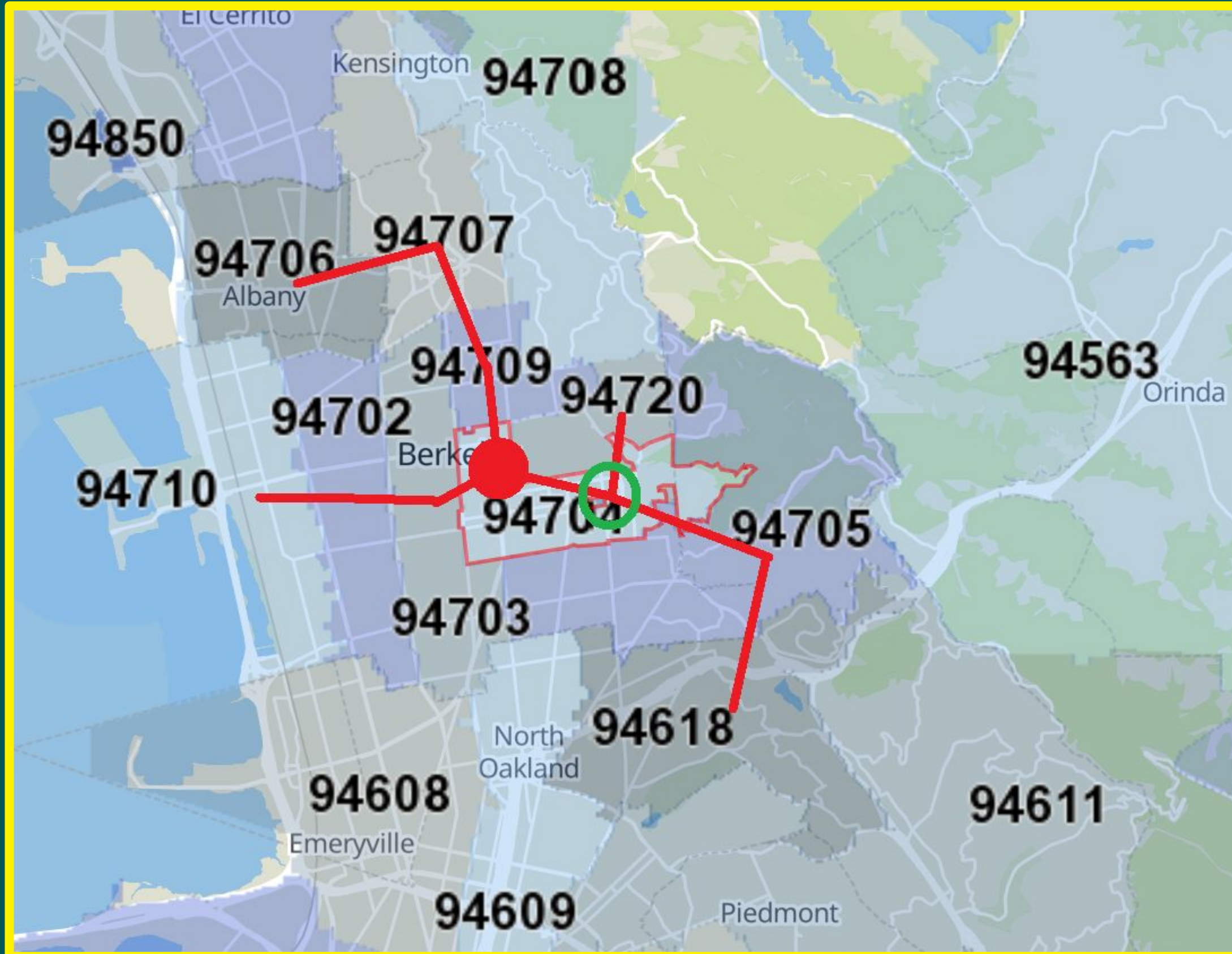
Minimum Spanning Tree – Results

Delivery within 2 miles of kitchen:

- 3 routes for drones and robots
- add non-BART pick up locations at intersections



Minimum Spanning Tree – Map



- **BART Station**
- **Intersection**
 - potential new pickup location



MongoDB

Document Database

- Using alternative methods of delivery
 - Drones
 - Electric Vehicles
- Accelerating customers logins

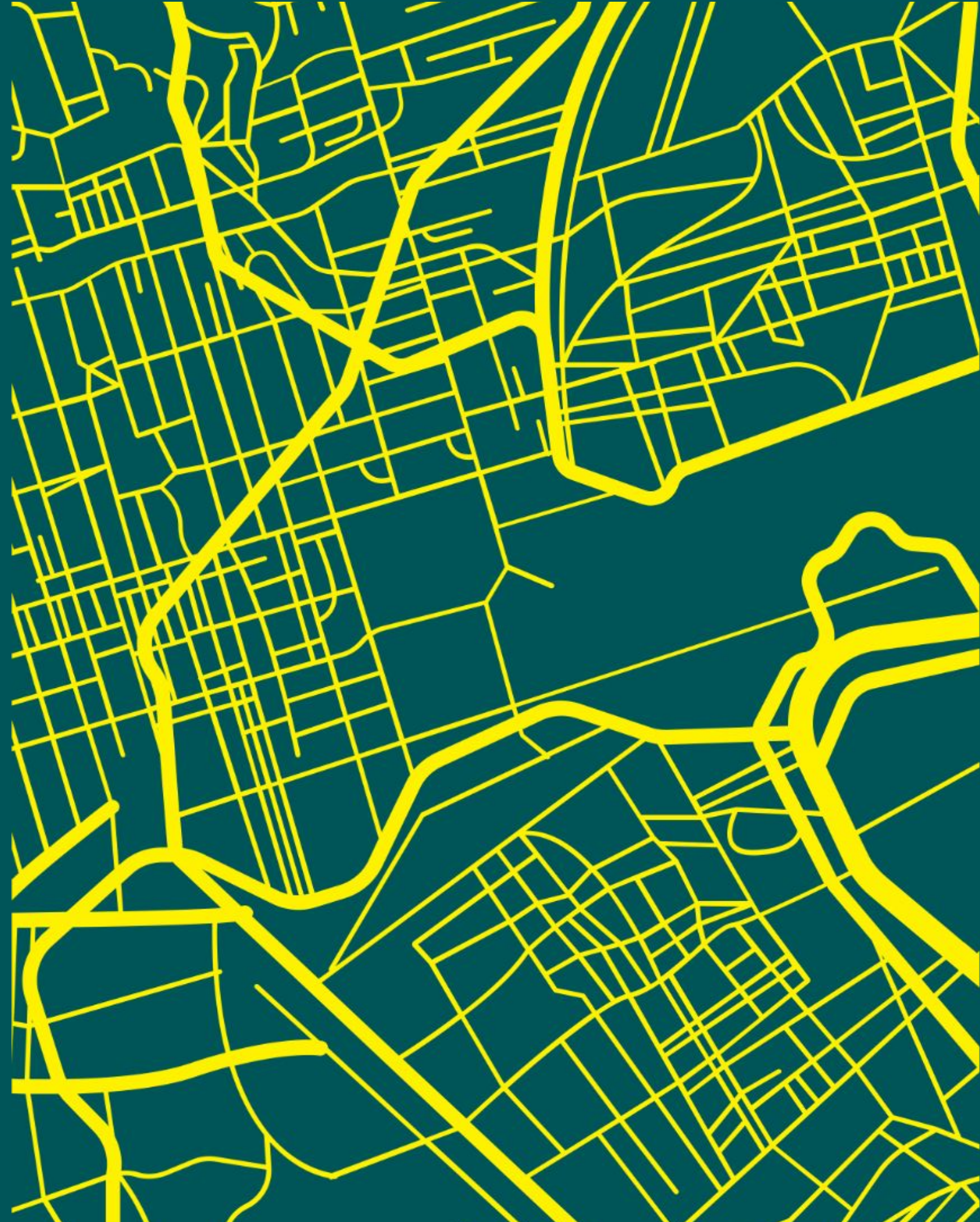
MongoDB – Business Cases



- Managing electric delivery vehicles in BART system
 - pre-compute shortest paths from kitchen, and store them
 - **mongoDB** – faster when querying + ability to scale up
- Live traffic report
 - POVs show stations, customers, products, and vehicles affected
 - **mongoDB** – flexible schema + stored data format
- Faster customer login
 - manage broad range of data operations
 - **relational DB** – pull from many tables → ↑ login time

Redis

- Real-Time Tracking of Delivery Drones and Robots
 - Continuously send coordinates and updates for routing and scheduling



Redis – Features



- **Pub / Sub, Lists, Sorted Sets**
 - real-time messaging
- **Geospatial Toolset**
 - query objects based on geography
- **In-Memory Database**
 - low latency, high throughput
- **Clustering**
 - scalability, data redundancy



Database – Drawbacks

- **Performance Issues with Real-Time Data**
 - traditional DB – not optimized
 - relational DB – struggle with volume
 - **Geospatial Queries are Complex**
 - relational DB – less efficient and need plugins
 - **Scalability Concerns**
 - relational DB – scaling → adding servers
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Thank you!